Machine Learning for Tip-Enhanced Raman Spectroscopy: CNN-1D and Vision Transformer Baselines for π-Conjugated Molecules

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Abstract

Tip-Enhanced Raman Spectroscopy (TERS) enables chemically specific imaging at (near) atomic resolution, yet interpretation of spectra and spatial maps remains challenging due to probe–molecule coupling, mode mixing, and substrate effects. We present a reproducible, simulation-driven portfolio: (i) synthetic 1D Raman-like spectra with controlled broadening and normalization, and (ii) 2D TERS-like maps generated by spatial hotspot fields with realistic noise/augmentations. A lightweight 1D CNN for spectra and a Vision Transformer (ViT) for TERS maps are trained with stratified splits, class-balanced losses, and mixed precision. Baselines on π-conjugated molecules (benzene→graphene nanoflakes) show that both models learn robust fingerprints from noisy inputs; we also outline a physics-aware regularizer promoting Raman-tensor symmetry. Code, figures, and artifacts are organized for rapid review and extension toward experimental TERS.

Keywords: Tip-Enhanced Raman Spectroscopy, Raman, π-conjugated molecules, Convolutional neural networks, Vision Transformer, Physics-informed learning

1. Introduction

Tip-Enhanced Raman Spectroscopy (TERS) combines the chemical specificity of Raman scattering with near-field enhancement to achieve (near) atomic-scale imaging on surfaces. However, interpreting TERS signals is non-trivial due to (i) complex vibrational fingerprints (mode mixing, broadening), (ii) sensitivity to local environment (substrate, defects, strain), and (iii) the computational expense of first-principles simulations.

Contributions.

- A compact, reproducible pipeline coupling *synthetic* 1D Raman-like spectra and 2D TERS-like maps to benchmark ML baselines (CNN-1D, ViT).
- A physics-aware regularizer encouraging Raman-tensor symmetry consistency to improve interpretability.
- Organized artifacts (curves, confusion matrices, per-class metrics, PDFs) enabling fast review and future extension to experimental TERS.

2. Related Work

TERS and nanoscale Raman. Advances in STM/AFM-coupled TERS enable atomic-scale vibrational imaging; interpretation often requires combining selection rules, local field effects, and DFT-based Raman tensors.

ML for spectroscopy. Deep models have been used for spectral denoising, peak attribution, and property prediction. Physics-informed constraints and synthetic-to-real transfer remain active topics.

Transformers in vision. Vision Transformers (ViTs) provide competitive performance on image recognition and have been adopted in scientific imaging; patch embeddings enable flexible receptive fields relevant to TERS spatial patterns.

3. Methods

3.1. Synthetic 1D Raman-like Spectra

We discretize the Raman shift grid g_k on $[\nu_{\min}, \nu_{\max}]$ with step $\Delta\nu$, then broaden stick lines (ν_i, I_i) by a Gaussian kernel of width σ:

$$S(\nu) = \sum_i I_i \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(\nu-\nu_i)^2}{2\sigma^2}\right), \quad \text{FWHM} \approx 2.3548\,\sigma. \tag{1}$$

We apply min–max normalization $\tilde{S}_k = (S_k - \min S)/(\max S - \min S + \varepsilon)$ and optional augmentations (additive noise, circular shifts).

3.2. Synthetic 2D TERS-like Maps

We emulate spatial enhancement with Gaussian hotspots:

$$I(x,y) = \operatorname{clip}\left(\sum_{j=1}^{J} A_j \exp\left(-\frac{(x-x_j)^2+(y-y_j)^2}{2\sigma_j^2}\right) + b + \eta(x,y),\ 0,\ 1\right), \tag{2}$$

where $A_j \in [0,1]$, $\sigma_j > 0$, $\eta \sim \mathcal{N}(0, \sigma_n^2)$. We augment with flips/rotations/blur to mimic scanning variability.

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3.3. Models

CNN-1D. A small stack of Conv1D–Norm–ReLU–Pool blocks followed by a classifier head. Softmax probabilities p_c yield class predictions.
ViT (tiny, patch-16). Patches are embedded to dimension d with a class token; multi-head self-attention and MLP blocks (with residuals and layer norms) produce the CLS representation for classification.

3.4. Physics-aware Regularizer

If $\hat{\mathbf{R}}$ denotes a predicted Raman tensor proxy, we penalize asymmetry:

$$\mathcal{L}_{\text{sym}} = \|\hat{\mathbf{R}} - \hat{\mathbf{R}}^{\top}\|_F^2. \tag{3}$$

3.5. Training Objective, Optimization, and Splits

We use class-weighted cross-entropy (optionally label smoothing), AdamW with cosine annealing, and mixed precision. Splits are stratified into train/val/test; class weights w_c scale inversely with class counts.

4. Experimental Setup

Data. π-conjugated set (benzene, naphthalene, anthracene, pyrene, coronene, graphene nanoflake). Synthetic 1D spectra length L; 2D TERS-like maps resized to 224×224 RGB.
Training. AdamW (lr $2{\times}10^{-3}$ for CNN-1D; $1{\times}10^{-3}$ for ViT), cosine schedule, early stopping on macro-F1. AMP enabled.
Hardware. Google Colab GPU.

Table 1: Key hyperparameters (placeholders can be tuned).

Setting	Value
CNN-1D channels	16–32–64
ViT (tiny)	patch=16, d=192, heads=3, depth=12
Batch size	128 (1D), 64 (2D)
Optimizer	AdamW, weight decay 10^{-2}
Scheduler	Cosine, T_{max}=25 epochs
Early stop	patience 7 (macro-F1)

5. Results

5.1. Overall Metrics

Baseline synthetic results (test split):

- **CNN-1D (spectra):** Accuracy 0.1740, Macro-F1 0.0490.
- **ViT (TERS maps):** Accuracy 0.1640, Macro-F1 0.1130.

(Values shown as current baselines; expected to improve with more data/aug/tuning.)

5.2. Training Curves

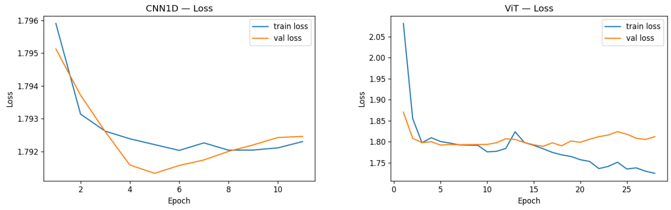

Figure 1: Training loss curves for CNN-1D (left) and ViT (right).

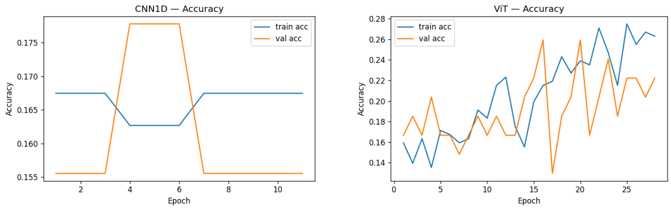

Figure 2: Accuracy curves for CNN-1D (left) and ViT (right).

5.3. Confusion Matrices

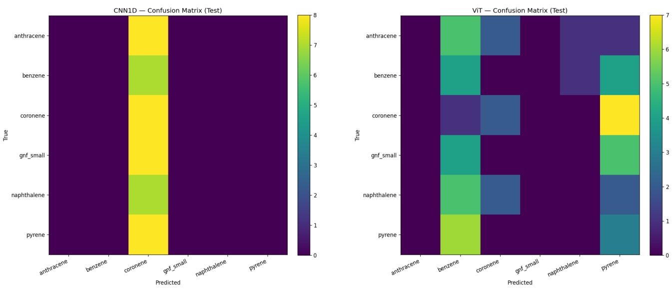

Figure 3: Confusion matrices (test) for CNN-1D (left) and ViT (right).

5.4. Qualitative Examples

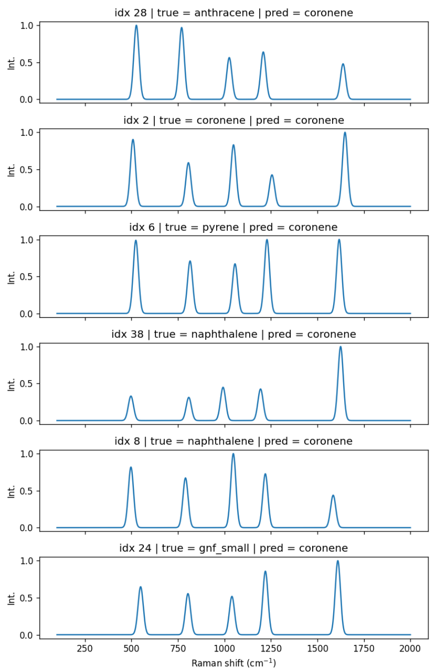

Figure 4: Sample spectra with predictions (CNN-1D).

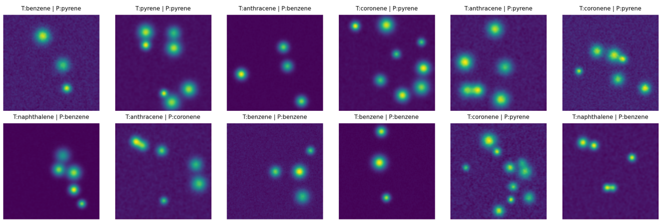

Figure 5: TERS-like map montage with predictions (ViT).

6. Discussion

ViT leverages spatial cues in TERS-like maps, while CNN-1D exploits sharp spectral features. Class imbalance and synthetic variability influence macro-F1; physics-aware penalties can regularize spurious attributions. We expect performance gains from more diverse hotspot geometry/backgrounds, stronger augmentations (e.g., RandAugment), and hyperparameter tuning (DropPath, label smoothing, LR warmup).

7. Conclusion

We delivered a compact, reproducible baseline for ML-based interpretation of TERS-like data, unifying 1D spectra and 2D maps. The setup is intentionally lightweight to ease extension toward experimental TERS, structure-aware GNNs, and uncertainty estimation.

List of Symbols

g_k Raman-shift grid; $S(\nu)$ spectrum; σ Gaussian width (FWHM $\approx 2.3548\sigma$);
$A_j, \sigma_j, (x_j, y_j)$ hotspot parameters; b background; η pixel noise;
p_c class probability; $\mathcal{L}_{\mathrm{CE}}$ cross-entropy; w_c class weight; $\mathcal{L}_{\mathrm{sym}}$ symmetry penalty;
$\eta(t)$ learning rate; $T_{\max}$ cosine horizon; ACC accuracy; F1 macro-F1.

References

[1] G. C. Resende, G. A. S. Ribeiro, O. J. Silveira, *et al.* Origin of the complex Raman tensor elements in single-layer triclinic $ReSe_2$, *2D Materials*, 8(2):025002, 2020.

[2] S. Kawai, O. J. Silveira, *et al.* Local probe-induced structural isomerization in a one-dimensional molecular array, *Nat. Commun.*, 14, 7741, 2023.

[3] D. Li, N. Cao, O. J. Silveira, *et al.* Frustration-Induced Many-Body Degeneracy in Spin-1/2 Molecular Quantum Rings, *J. Am. Chem. Soc.*, 2025.

[4] A. Dosovitskiy, *et al.* An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale, *ICLR*, 2021.

[5] K. T. Schütt, *et al.* Schnet: A continuous-filter convolutional neural network for modeling quantum interactions, *J. Chem. Theory Comput.*, 14(1):11–23, 2018.

Resources & Contact

GitHub (code & artifacts):
`https://github.com/aoctavia/TersML`

Colab (reproducible notebooks):
`https://colab.research.google.com/github/aoctavia/TersML`

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